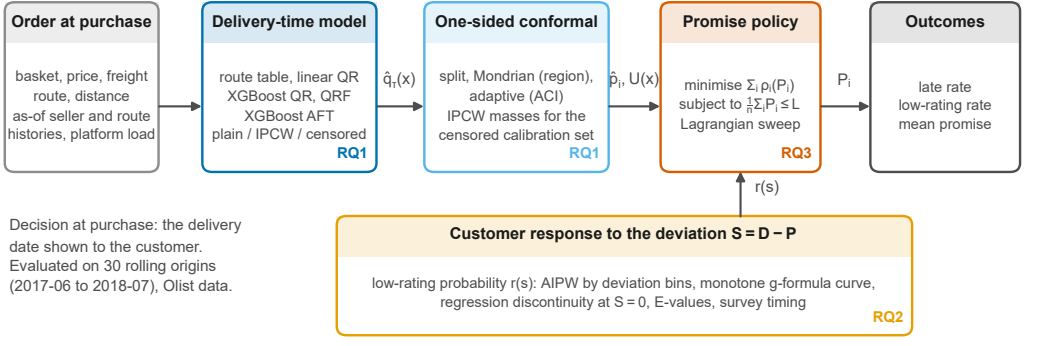

Order at purchase
basket, price, freight
route, distance
as-of seller and route
histories, platform load
Delivery-time model
route table, linear QR
XGBoost QR, QRF
XGBoost AFT
plain / IPCW / censored
RQ1
$\hat{q}_\tau(x)$
One-sided conformal
split, Mondrian (region),
adaptive (ACI)
IPCW masses for the
censored calibration set
RQ1
$\hat{p}_i$, U(x)
Promise policy
minimise $\Sigma_i p_i(P_i)$
subject to $\frac{1}{n}\Sigma_i P_i \le L$
Lagrangian sweep
RQ3
P_i
Outcomes
late rate
low-rating rate
mean promise
r(s)
Customer response to the deviation S = D − P
low-rating probability r(s): AIPW by deviation bins, monotone g-formula curve,
regression discontinuity at S = 0, E-values, survey timing
RQ2
Decision at purchase: the delivery
date shown to the customer.
Evaluated on 30 rolling origins
(2017-06 to 2018-07), Olist data.